Master's Thesis

Multi-Agent Reinforcement Learning for Joint Interference Avoidance and Gateway Load Balancing via Beam Hopping in NTN

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감사의 글

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Abstract

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List of Abbreviations

3GPP	3rd Generation Partnership Project
ACK	Acknowledgment
BH	Beamhopping
BHO	Basic Handover
CHO	Conditional Handover
DDQN	Double DQN
DNN	Deep Neural Network
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
ECEF	Earth-centered, Earth-fixed
EIRP	Effective Isotropic Radiated Power
FLSO	Feeder Link Switchover
FSPL	Free Space Path Loss
GW	Gateway
HO	Handover
IIR	Infinite Impulse Response
KPI	Key Performance Indicator

L3	Layer 3
LEO	Low Earth Orbit
LoS	Line-of-Sight
MDP	Markov Decision Process
MTS	Minimum ToS
NLoS	Non-Line-of-Sight
NP	Non-deterministic Polynomial time
NTN	Non-Terrestrial Networks
PER	Prioritized Experience Replay
PPO	Proximal Policy Optimization
RACH	Random Access Channel
RB	Resource Block
RLF	Radio Link Failure
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SNR	Signal-to-Noise Ratio
SINR	Signal-to-Interference-plus-Noise Ratio
TD	Temporal-Difference
TN	Terrestrial Networks
ToS	Time of Stay
UE	User Equipment
UHO	Unnecessary Handover
UFLSO	Unnecessary Feeder Link Switchover

Chapter 1. Introduction

1.1. Background and motivation

LEO satellite networks have gained significant attention in recent years due to their potential to provide global broadband coverage with low latency. Operating at altitudes between 160 km and 2,000 km above the Earth's surface, LEO satellites offer several advantages over GEO satellites, including reduced signal propagation delay and lower power requirements for communication [1]. LEO satellite networks are also critical for military communications, where reliable and low-latency connectivity is essential for tactical operations, such as real-time battlefield coordination control [2].

LEO satellites are categorized into transparent types, connecting terrestrial gNBs and GWs to UEs, and regenerative types with onboard gNBs. However, the high orbital velocity of both LEO satellites, approximately 7–8 km/s, introduces significant challenges in maintaining stable communication links such as feeder link and service link. For almost transparent types, LEO satellite feeder link operating with a break-before-make policy must perform switchovers as they pass over multiple

ground GWs, which in turn induces HO for the UEs receiving LEO satellite services because the gNB associated with the each GW [3]. In the 6G mMTC scenarios, HO failures may occur due to the transmission of HO commands to a large number of UEs. These frequent FLSO and UE HO can lead to service quality degradation and increased latency, posing a major challenge for NTN in the context of 6G communications [4]. According to [5], UE power consumption during HOs ranges from 100 mW in the best case to 400 mW in the worst case. This energy overhead significantly impacts low-power battlefield devices, including IoT sensors, UAVs, and robots, hastening battery depletion and degrading network energy efficiency. Frequent FLSO and HO can disrupt mission-critical communications, increasing latency and risking security breaches in contested environments. Consequently, optimizing the FLSO strategy to minimize UE HO is crucial for ensuring seamless service in NTN.

1.2. Contributions

In this paper, we propose a reinforcement learning-based FLSO scheme that optimizes HO for UEs in LEO satellite coverage. The main contributions are summarized as follows:

- I created a dynamic system environment to facilitate the reinforcement learning agent's training process. By leveraging locally observable metrics, such as received signal strength and LEO satellite ephemeris data, the agent adapts to rapid orbital dynamics, thereby enhancing link stability and reducing service outages.

- I designed a reward function leveraging RSRP and ToS to optimize long-term LEO satellite and UE connectivity and optimize FLSO, UFLSO and UE HO in LEO satellite networks.

- I proposed a dueling DQN-based FLSO algorithm to enhance robustness in noisy and dynamic LEO environments for transparent LEO satellites. This algorithm dynamically selects ground GW to reduce UE HO frequency, enhance RSRP, and extend ToS across varying UE distributions.

1.3. Overview

The remainder of this thesis is structured as follows: Sec. II discusses handover, feeder link switching, and related research in beam hopping and resource allocation. Sec. III presents the proposed reinforcement learning framework in detail. Sec. IV discusses the simulation results and compares the proposed strategy with conventional approaches. Finally, Sec. V concludes the thesis.

Chapter 2. Related works

2.1. LEO satellite HO and FLSO schemes

In an NTN environment, the continuity of the service link between the UE and the satellite is ensured through HO. HO is the procedure in Fig. 2.1 by which a UE changes the serving cell. Generally, this network-controlled procedure consists of three main phases: preparation, execution, and completion. Initially, the source gNB configures the UE to perform measurements on neighboring cells, and the UE sends a MR back to the network. Based on this report and other RRM information, the source gNB decides to initiate a HO and sends a HO REQUEST to a target gNB. The target gNB performs admission control, prepares resources, and responds with a HO REQUEST ACK. The source gNB then commands the UE to switch to the target cell by sending an RCR message. Upon receiving this, the UE synchronizes with the target cell and sends an RCRC message to finalize the process. Subsequently, the target gNB requests the core network to switch the data path and triggers the release of the UE's resources at the source gNB. However, applying this procedure in NTNs presents unique challenges. Unlike in terrestrial net-

works, the difference in received signal strength between satellites is often small, making traditional measurement-based triggers less reliable. Consequently, distance- or time-based HO criteria are also employed to manage the predictable mobility of satellites. In addition to the service link, the continuity of the feeder link connecting the satellite to the ground GW is equally important. The presence or absence of a gNB onboard the satellite determines whether a feeder link switchover impacts the service link. For a transparent satellite without an onboard gNB, a feeder link switchover changes the serving GW on the ground, which in turn changes the serving gNB, resulting in a necessary HO procedure for the UE. In contrast, for a regenerative satellite equipped with an onboard gNB, a feeder link switchover can be handled at the transport layer without changing the serving gNB, making the switch transparent to the UE and avoiding an HO.

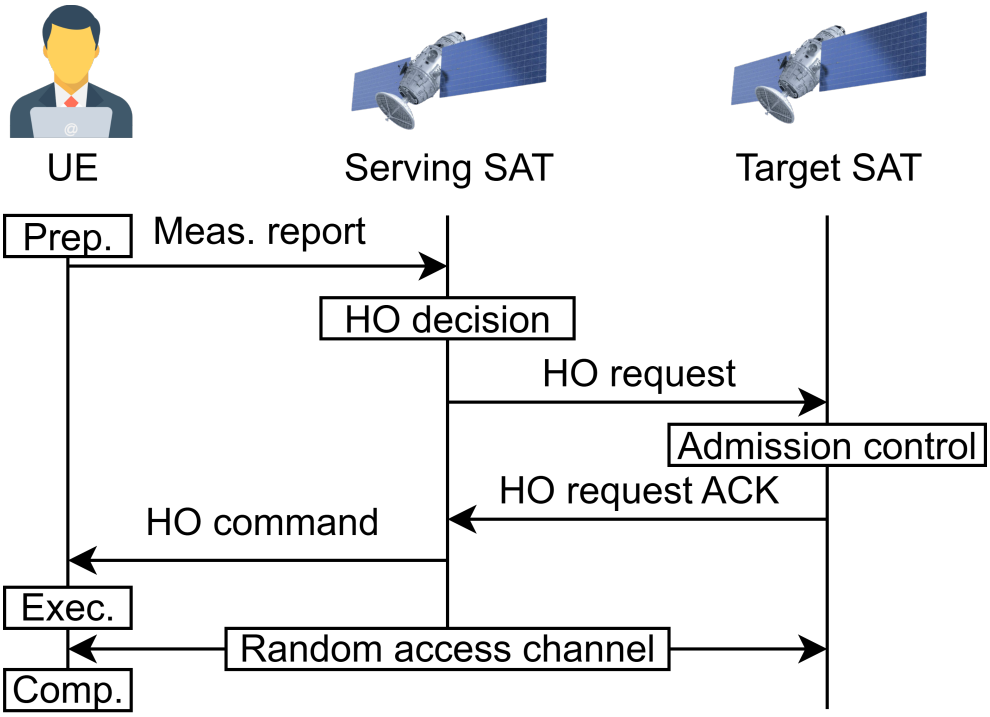


Fig. 2.1. Basic HO procedure

2.2. Reinforcement learning-based Beam hopping

BH has been extensively studied as a key resource-allocation technique to cope with spatially and temporally heterogeneous ground traffic. In LEO satellite systems, multi-satellite cooperation becomes a central challenge due to overlapping coverage and rapidly varying link conditions.

Lin *et al.* [6] proposed a framework that decomposes the multi-satellite BH problem into two core components: load balancing and interference avoidance. Exploiting the multiple-coverage property of LEO constellations, the framework first distributes hotspot traffic across several satel-

lites load balancing. Each satellite then independently designs a BH pattern for its assigned cells to satisfy per-cell demand while maintaining spatial isolation among simultaneously illuminated beams. Finally, the sequence of per-satellite BH patterns is jointly scheduled to mitigate inter-satellite interference avoidance. While this decomposition provides a clear problem structure, the approach in [6] relies on predefined heuristics to solve the subproblems sequentially, limiting adaptability to non-stationary traffic and unpredictable network dynamics.

Complementary to the above line, Xie *et al.* propose a DRL approach for multi-LEO BH with joint beam scheduling and power allocation [7]. Their method formulates a hybrid action space—discrete beam-illumination patterns and continuous per-beam power—and employs PPO with two policy heads sharing a common feature extractor. The objective jointly optimizes long-term throughput and long-term cumulative average delay under full-frequency reuse, explicitly accounting for both intra- and inter-satellite interference in the SINR model. Simulation results show notable gains over heuristic baselines (e.g., up to 8.9% higher throughput and up to 69.2% lower LTCAD), highlighting the potential of DRL for traffic-driven BH in time-varying environments. Nonetheless, the learning agent is effectively centralized and assumes access to global network state (e.g., via NOCC), and the model does not explicitly consider GW selection or feeder-link switchover effects on service continuity.

To address these limitations, this paper adopts a multi-agent reinforcement learning MARL formulation. We model each satellite as an

autonomous agent that observes local network states (e.g., per-cell demand, channel/interference conditions, and GW status) and coordinates with other agents to learn a cooperative BH policy. The learned policy simultaneously optimizes interference avoidance and gateway load balancing, thereby improving service SINR and alleviating traffic bottlenecks. In contrast to the heuristic decomposition in [6] and the centralized DRL paradigm in [7], our MARL framework natively supports distributed decision-making under partial observability and explicitly couples BH with GW selection and feeder-link switchover, which are critical for end-to-end performance in practical NTN deployments.

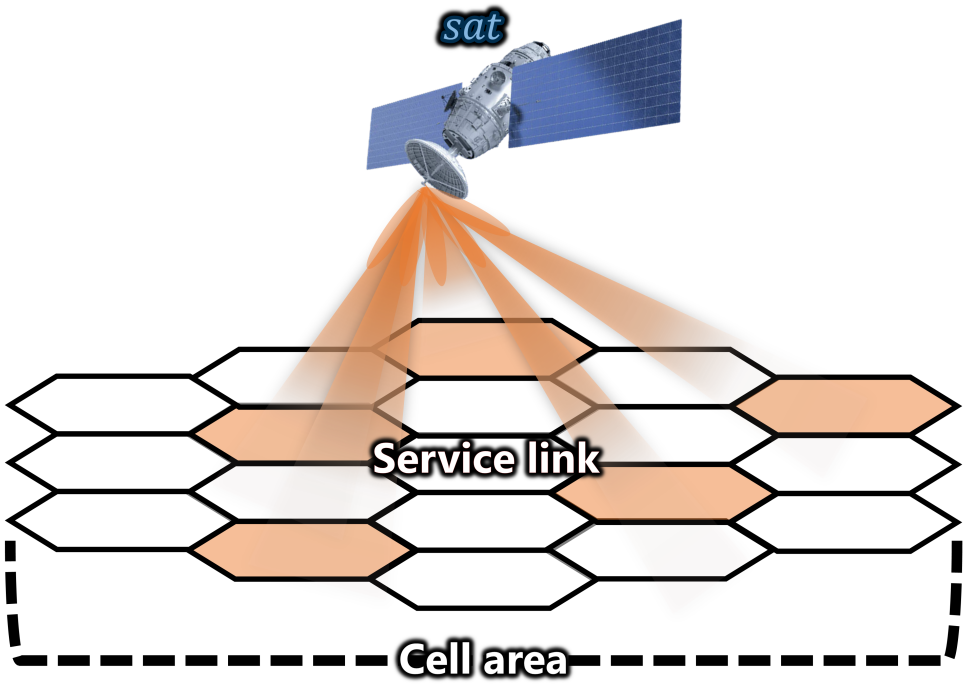


Fig. 2.2. Satellite beam hopping system

Chapter 3. Proposed reinforcement learning-based link scheduling algorithm

3.1. System model

This thesis considers the transparent LEO satellite constellation consist of N LEO satellites, represented by the set $\mathcal{S} = \{n \mid n = 1, 2, \ldots, N\}$. The LEO satellite connects to ground GW via feeder link, where the set of GW denoted as $\mathcal{G} = \{m \mid m = 1, 2, \ldots, M\}$. The GWs consists of both an antenna and a gNB. Since the GWs are widely separated, each is equipped with its own dedicated gNB. The coordinates of each LEO satellite in the ECEF coordinate system at a given time slot t, where $t \in \{0, 1, \ldots, T\}$, are given by $l_n(t) = (x_n(t), y_n(t), z_n(t))$. As the LEO satellite services the region, it covers UEs, denoted by the set $U = \{u_t \mid u_t = N_t(\mu, \sigma^2)\}$, distributed according to a 2-D Gaussian footprint. With a hard FLSO applied, each UE within the LEO satellite's coverage area simultaneously initiates a HO to the subsequent gNB upon every FLSO. Fig. 3.1 shows the network scenario where a transparent Starlink LEO satellite passes over M ground GWs in the United States, serving UEs.

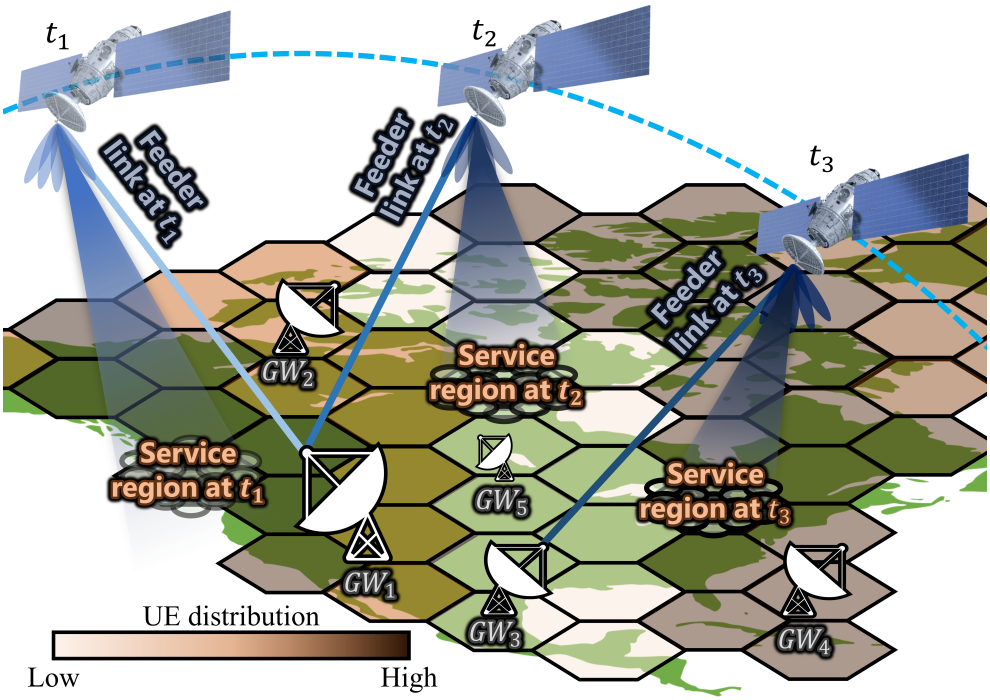


Fig. 3.1. Network scenario of a LEO satellite performing FLSO while serving UEs across Starlink GWs in the United States.

The LEO satellite n evaluates the RSRP from signals transmitted by the GW m at periodic interval T_m. The RSRP $R_{m,n}$ is calculated as follows:

$$R_{m,n} = EIRP_m + G_n - PL_{m,n} - 10\log_{10}(q_{RB}), \tag{3.1}$$

where $EIRP_m$ is the EIRP of the GW m, defined as $EIRP_m = P_m + G_m$, with P_m being the transmit power and G_m the transmit antenna gain of the GW m towards the LEO satellite n. G_n represents the receive antenna gain of the LEO satellite n. The term $10\log_{10}(q_{RB})$ is used to normalize the RSRP measurement, accounting for a single resource block out of

the total resource blocks q_{RB} allocated to the link. $PL_{m,n}$ is the channel model between the GW m and the LEO satellite n is provided in the 3GPP documentation [8] and is calculated as follows:

$$PL_{m,n} = P_{LOS,m,n}(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n}) + P_{NLOS,m,n}(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n} + CL_{m,n}), \quad (3.2)$$

$P_{LOS,m,n}$ and $P_{NLOS,m,n}$ denote the probabilities that the LEO satellite n to GW m is in a LOS and NLOS condition, respectively. $FSPL_{m,n}(d_{m,n}, f_m)$ denotes the FSPL as a function of the distance $d_{m,n}$ between the LEO satellite n and the GW m and the frequency f_m of the GW m uplink feeder link frequency. $SF_{m,n}$ represents the shadow fading, which accounts for the signal attenuation caused by obstacles such as buildings, trees, or terrain variations in the propagation path. $CL_{m,n}$ denotes the clutter loss, reflecting additional signal degradation due to environmental clutter. Specifically, the $FSPL_{m,n}$ is calculated as follows:

$$FSPL_{m,n} = 32.45 + 20log_{10}(f_m) + 20log_{10}(d_{m,n}). \quad (3.3)$$

At each measurement instant t, the LEO satellite n samples the RSRP from GW m and all neighboring GWs within instantaneous visibility region, yielding the raw RSRP $R_{m,n}(t)$. To mitigate the effects of shadow fading and measurement noise, these raw RSRP $R_{m,n}(t)$ samples are processed at the RRC layer using a L3 IIR filter [9]. Denoting the filtered

RSRP by $\widetilde{R}_{m,n}(t)$, the update equation is calculated as follows:

$$\widetilde{R}_{m,n}(t) = aR_{m,n}(t) + (1-a)\widetilde{R}_{m,n}(t), \tag{3.4}$$

where $a \in (0,1)$ is the L3 forgetting factor, defined as $a = \left(\frac{1}{2}\right)^{K/4}$, and K is the filter coefficient. The initial condition is set as $\widetilde{R}_{m,n}(0) = R_{m,n}(0)$. Finally, the LEO satellite compiles a measurement report containing the L3-filtered RSRP $\widetilde{R}_{m,n}(t)$ of the serving and neighboring GWs and transmits this report back to the serving GW m over the feeder link.

To quantify the link quality during the simulation, we calculate the channel capacity C using the Shannon capacity formula, which is calculated as follows:

$$C = B_m \log_2(1+\zeta), \tag{3.5}$$

where the B_m is the uplink bandwidth of GW m and ζ is the SNR calculated as follows:

$$\zeta = EIRP_m + \frac{G_n}{T} - k - PL_{m,n} - B_m, \tag{3.6}$$

where, k is the Boltzmann constant, G_n/T is the antenna gain-to-noise temperature ratio, is calculated as follows:

$$\frac{G_n}{T} = G_n - N_f - 10\log_{10}\left(T_0 + (T_a - T_0)10^{-0.1N_f}\right), \tag{3.7}$$

where N_f is LEO satellite noise figure, T_0 and T_a is ambient and antenna temperature.

3.2. Problem formulation

In this section, we address the optimization problem of performing optimal FLSO to minimize the number of HO for UE served by a LEO satellite. We formulate this problem as an NP-hard problem [10] and propose a solution by modeling it as a MDP, which is then solved using reinforcement learning techniques.

In LEO networks, frequent FLSO and HO events cause elevated latency and power consumption. FLSO $FS_{m,n}$ and HO $H_{m,n}$ are measured when the serving m transitions from $t-1$ to t calculated as follows:

$$FS_{m,n} = \sum_{t=0}^{T} \left(1 \text{ if } m_t \neq m_{t-1} \text{ else } 0\right), \tag{3.8}$$

$$H_{m,n} = \sum_{t=0}^{T} \left(u_{t-1} \text{ if } m_t \neq m_{t-1} \text{ else } 0\right). \tag{3.9}$$

To address these issues, we formulate an optimization problem that enables the LEO satellite to perform optimal FLSO while minimizing UE HO and UFLSO, and maximizing ToS and RSRP, meeting the conditions

calculated as follows:

$$\textbf{P1:} \quad \min_{m} \sum_{n=1}^{N} \sum_{m=1}^{M} \sum_{t=0}^{T} \left(FS_{m,n} + H_{m,n} + UFS_{m,n} - ToS_{m,n} - \widetilde{R}_{m,n} \right), \tag{3.10}$$

s.t.

$$\textbf{C1}: \quad \theta_{m,n} \geq \theta_{\min}, \quad \forall n, \forall m, \forall t, \tag{3.11}$$

$$\textbf{C2}: \quad \widetilde{R}_{m,n} \geq R_{\min}, \quad \forall n, \forall m, \forall t, \tag{3.12}$$

$$\textbf{C3}: \quad I_{m,n}(t) = 1, \quad \forall n, \forall m, \forall t. \tag{3.13}$$

In **P1**, $UFS_{m,n}$ represents the cost of an UFLSO, which occurs when the LEO satellite n switches to another GW within the MTS and $ToS_{m,n}$ represents ToS the duration of the LEO satellite n remains connected to the GW m. The objective function minimizes the total cost, which includes the FLSO, HO of UEs and UFLSO events and maximizes ToS and RSRP across N LEO satellites, M GWs, and T time slots. Constraint **C1** ensures the elevation angle $\theta_{m,n}$ between LEO satellite n and GW m exceeds the minimum threshold $\theta_{\min}$, guaranteeing reliable communication by maintaining a sufficient line-of-sight. Constraint **C2** ensures the filtered RSRP $\widetilde{R}_{m,n}$ exceeds the minimum required rate $R_{\min}$, ensuring adequate link quality for effective data transmission. Constraint **C3** enforces that each LEO satellite connects to exactly one GW at all time t, using the indicator variable $I_{m,n}(t)$, which is 1 if LEO satellite n is connected to GW m at time t.

3.3. Proposed algorithm

To address the problem, we transform it into a dynamic programming framework to derive an optimal FLSO policy. We model the FLSO optimization problem as an MDP due to its ability to handle sequential decision-making in dynamic environments.

3.3.1 MDP modeling

The LEO satellite's state s_t evolves over time and actions a_t impact future reward r_t with discount factor γ [11]. This framework enables the LEO satellite to learn an optimal FLSO policy through reinforcement learning, effectively balancing service continuity and HO minimization. The state space s_t of the MDP is calculated as follows:

$$s_t = \left\{ l_n(t), m_t, \widetilde{R}_{m,n}(t), u_t \right\}. \tag{3.14}$$

The state is defined as a tuple comprising the LEO satellite's ECEF coordinates $l_n(t)$, the currently connected GW m_t, the RSRP of the current GW $\widetilde{R}_{m,n}(t)$, and the number of UEs currently being served u_t at time t, configured to be accessible and interpretable by the LEO satellite.

The action space a_t involves the LEO satellite selecting the optimal GW at each step. Accordingly, our action space is calculated as follows:

$$a_t = \{m_t | m_t \in \mathcal{G}_t\}. \tag{3.15}$$

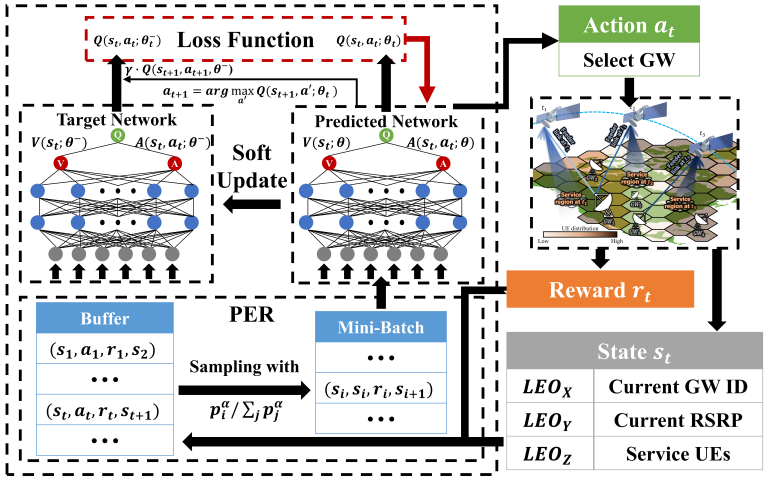


Fig. 3.2. The architecture of the proposed dueling DQN for FLSO optimization in LEO satellite networks.

The action comprises the set of GW $\mathcal{G}_t$ denotes the set of all available GW at t.

The immediate reward r_t is formulated to jointly promote service continuity and signal quality while penalizing both necessary and UFLSO. It is calculated as follows:

$$r_t = (1 + \rho u_t)FS(t) + UFS(t) - \Big(ToS(t) + \nu \widetilde{R}_{m,n}(t)\Big). \tag{3.16}$$

The parameters ρ and ν are normalization factors used to align the values with other metrics.

Discount factor γ is a value between 0 and 1, which represents the rate at which future rewards are discounted when considering them at the current step.

3.3.2 Proposed dueling DQN algorithm

In this paper, we employ a reinforcement learning algorithm to enable LEO satellites to operate in an dynamic environment. As depicted in Fig. 2, the architecture of the proposed dueling DQN algorithm, incorporating a ϵ-greedy policy, is designed to optimize FLSO in LEO satellite networks. The LEO satellite receives feedback to execute actions that sustain both a prolonged ToS and an elevated RSRP, while minimizing the frequency of FLSO, UFLSO, and UE HO events. We enhance the traditional DQN framework by incorporating DDQN, PER, and dueling DQN to improve stability, learning efficiency, and robustness against noise in LEO satellite communication systems.

The DQN algorithm [12] approximates the Q-function $Q(s_t, a_t; \theta)$, which maps a state s_t and action a_t to an expected cumulative reward, using a DNN parameterized by θ. Two DNNs are utilized: an online network for action selection and a target network for value estimation. The Q-function is trained to minimize the following loss function and loss function is calculated as follows:

To improve robustness against random noise, such as channel shadow fading and variations in UE distribution, we utilize a dueling DQN [13]. The dueling DQN architecture enhances the traditional DQN by splitting the Q-function into two separate streams: the state-value function $V(s; \theta)$ and the advantage function $A(s, a; \theta)$. The Q-function is then calculated

Algorithm 3.1 Dueling DDQN-based Algorithm for UE HO

Initialization: Random weights θ, replay memory $\mathcal{R}$ with capacity $\mathcal{D}$, minibatch size $\mathcal{Z}$, total episodes $\mathcal{E}$, prioritization parameter α, importance sampling parameter β, exploration rate ϵ, Polyak averaging parameter τ
Input: State, action, discount factor
Output: Optimal FLSO decision
for *Episode* $= 1$ to $\mathcal{E}$ **do**
    Initialize the integrated environment
    for $t = 0$ to T **do**
        $a_t = \begin{cases} \arg\max_{a'} Q(s_t, a'; \theta) & \text{with } 1 - \epsilon \\ extrandomaction & \text{with } \epsilon \end{cases}$
        Execute action, observe reward and next state
        Store sample (s_t, a_t, r_t, s_{t+1}) into $\mathcal{R}$
        if $\mathcal{R} > \mathcal{D}$ **then**
            for $i = 1$ to $\mathcal{Z}$ **do**
                Sample (s_i, a_i, r_i, s_{i+1}) from $\mathcal{R}$ with probability p_i^α
                Calculate target network
            end for
            $\theta^- \leftarrow \tau \cdot \theta + (1 - \tau) \cdot \theta^-$
        end if
    end for
end for

as follows:

$$Q(s,a;\theta) = V(s;\theta) + \left(A(s,a;\theta) - \frac{1}{|A|} \sum_{a'} A(s,a';\theta) \right), \tag{3.17}$$

where $V(s;\theta)$ represents the value of the state s, $A(s,a;\theta)$ represents the advantage of taking action a in state s, and $|A|$ is the number of possible actions. This decomposition allows the network to better distinguish the value of a state from the advantage of each action, improving robustness against noise and enhancing learning efficiency in complex LEO satellite environments.

To further enhance the learning efficiency, we incorporate PER as proposed in [14]. PER prioritizes experiences with higher TD errors, where the TD error δ_i for transition i is defined as the difference between the target and current Q-values. The priority is calculated as follows:

$$p_i = |\delta_i| + \epsilon_p, \tag{3.18}$$

where $\delta_i = r_i + \gamma Q(s_{i+1}, a_{i+1}; \theta^-) - Q(s_i, a_i; \theta)$ is the TD error, and ϵ_p ensures non-zero priorities. PER uses $\alpha \in [0,1]$ to control sampling probability p_i^α and $\beta \in [0,1]$ to adjust importance sampling weights $(\mathcal{D} \cdot P(i))^{-\beta}$, where $\mathcal{D}$ is replay buffer limit and $P(i)$ is the probability of sampling transition i. This approach accelerates learning by focusing on significant transitions, which is beneficial in LEO satellite networks with dynamic and noisy conditions.

By combining dueling DQN with DDQN and PER, our approach

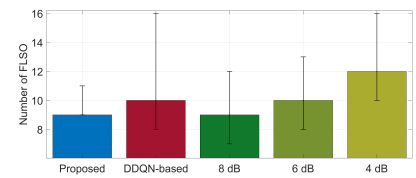
effectively addresses overestimation bias, accelerates learning, and mitigates the effects of random noise, leading to improved LEO satellite HO performance with prolonged ToS, elevated RSRP, and reduced FLSO, UFLSO, and UE HO events. The detailed process of the training is described in Algorithm 1.

Chapter 4. Performance evaluation

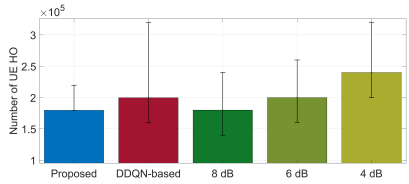
4.1. Simulation setup

In this section, we evaluate the performance of the proposed algorithm, DDQN-based FLSO and the signal-based FLSO algorithm through extensive simulations. The simulation environment is based on the actual deployment of SpaceX's Starlink GWs in the United States, servicing UEs in a mMTC environment. We consider operational GW sites and the LEO satellite's orbit was analyzed using the TLE data of Starlink No. 1008, retrieved on December 5, 2024. The LEO satellite periodically measures the RSRP of GWs as it passes over them. The simulation environment parameters are summarized in Table 4.1 and was performed in Python, iterating 1,000 times to analyze LEO satellite behavior and performance. The discount factor is set to 0.99 to further reward the future. Other algorithms are summarized as follows:

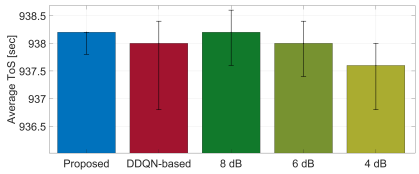
- The DDQN-based FLSO: It is trained using the identical state, action, and reward as defined for the dueling DQN. Both algorithms are initially trained with a UE spatial dispersion of 100, and the subsequent simulations are conducted using the learned Q-



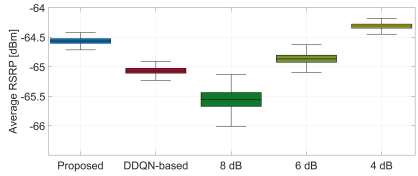
(a) Number of FLSO



(b) Number of UE HO



(c) Average ToS [sec]



(d) Average RSRP [dBm]

Fig. 4.1. Performance comparison of the proposed, DDQN-based, and signal-based FLSO algorithms across different metrics

networks. Learning proceeds via temporal-difference updates with experience replay and periodic target-network synchronization without dueling network.

- Signal-Based FLSO: It is implemented by adapting the conventional HO mechanism. Specifically, we employ the widely used A3 event-based FLSO algorithm as a baseline. In this algorithm, a FLSO event is triggered when the difference between the RSRP of the serving GW and that of target GWs exceeds a predefined threshold, denoted by $offset$.

Table 4.1. Simulation parameters

Number of LEO satellite N	1
Number of GW M	46
Carrier frequency f_m	28 [GHz]
Bandwidth B_m	480 [MHz]
GW Tx power P_m	47 [dBm]
GW Tx antenna gain G_m	49.5 [dBi]
LEO satellite Rx antenna gain G_n	38.5 [dBi]
Ambient temperature T_0	150 [K]
Antenna temperature T_a	290 [K]
Noise figure N_f	3 [dB]
Boltzmann constant k	1.379×10^{-23} [J/K]
Number of Resource Block q_{RB}	6
LEO satellite altitude h	530 [km]
UE distribution u_t	$\mu = 10^5$, $\sigma \in [100 \sim 5{,}000]$
Minimum elevation angle $\theta_{\min}$	25 [°]
Minimum RSRP of GW $R_{\min}$	-110 [dBm]
FLSO Parameter	**Value**
Period of measurement T_m	200 [ms]
Minimum time-of-stay	1 [s]
L3 filter forgetting factor a	$(\frac{1}{2})^{K/4}$
L3 filter coefficient K	4
A3 event $offset$	4, 6, 8 [dB]
Reinforcement Learning Parameter	**Value**
Learning rate	0.0001
Discount factor γ	0.99
Replay buffer limit $\mathcal{D}$	10^5
Batch size $\mathcal{Z}$	512
Total step per episode	4,700
Polyak tau τ	0.01
PER α	0.6
PER β Start	0.4

4.2. Key Performance Indicators

4.3. Results and Discussion

This is result.

Chapter 5. Conclusion

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